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Formalizing Explanation Design Through Interaction Patterns in Human-AI Decision Support

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Abstract. Trust in AI is crucial for effective and responsible use in high-stakes sectors like healthcare and finance. One of the most commonly used techniques to mitigate mistrust in AI and even increase trust is the use of Explainable AI models, which enables human understanding of certain decisions made by AI-based systems. Interaction design, the practice of designing interactive systems, plays an important role in promoting trust by improving explainability, interpretability, and transparency, ultimately enabling users to feel more in control and confident in the system's decisions. This paper introduces, based on an empirical study with experts from various fields, the concept of Explanation Stream Patterns, which are interaction patterns that structure and organize the flow of explanations in decision support systems. Explanation Stream Patterns formalize explanation streams by incorporating procedures such as progressive disclosure of explanations or interacting with explanations in a more deliberate way through cognitive forcing functions. We argue that well-defined Explanation Stream Patterns provide practical tools for designing interactive systems that enhance human-AI decision-making.

Keywords. Explainable Artificial Intelligence, Decision Support Systems, Interaction Design, Human-Computer Interaction

1. Introduction

Decision support systems (DSSs), particularly those that incorporate eXplainable AI (XAI), in high-stakes sectors like healthcare and finance, may improve decision quality and speed. However, these decisions typically affect individuals [1,2] and can be clas-

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sified as ”high-risk” systems under the AI act. Consequently, effective design of DSSs must go beyond technical robustness to promote user understanding, enable meaningful human-system interaction, and support human oversight [3].

Effective design alone is not enough, effective use in practice requires that DSSs are trustworthy [4]. In the XAI literature, the relation between trust and explainability is a central theme. Ali et al. [5] developed a unified perspective on the key concepts. In this perspective, *explainability* relies on *interpretability*, which builds user *trust* by making the system's decision-making process *transparent*. Trust, in turn, is reinforced by *robustness* and *reliability*, fostering confidence in the system's outputs. Furthermore, *interactivity* — a cornerstone of *interpretability* — enables users to actively engage with the system, deepening their understanding and trust.

Despite these advances, many current approaches to explainability rely on static, one-size-fits-all explanations that fail to account for the diversity of user needs [6]. Static explanations risk overwhelming users with excessive information, triggering cognitive overload and reliance on heuristics, which can lead to biased decisions. This underscores the importance of dynamic, user-tailored explanations that adapt to individual requirements and foster more effective decision-making. We argue that interaction design patterns offer a promising framework for addressing this challenge. Patterns encapsulate best practices for recurring design problems, providing adaptable, reusable guidelines that align high-level design principles with practical interface elements [7].

In the context of explainability, these patterns can help structure information delivery, quantify — or possibly mitigate — cognitive biases, improving decision accuracy [6]. For example, progressive disclosure patterns can prevent information overload by revealing explanations incrementally [8]. For example, patterns related to progressively disclosing explanations rather than presenting all information at once could help prevent information overload, allowing users to process data more effectively and make fewer errors [8]. Likewise, patterns that encourage reflective thinking — prompting users to question their initial judgments or engage with conflicting information — can mitigate confirmation bias and framing effects [9].

Beyond their immediate role in structuring explanations, interaction patterns also offer cross-context re-usability and a framework for systematic evaluation. Existing work on bias mitigation strategies and interaction by Bertrand et al. [10] identified three categories of interactivity that can serve as guiding principles: select, mutate, and dialogue with. For instance, *select* can be supported by interaction patterns such as filtering interfaces or contextual highlighting, helping users focus on the most relevant data. The *mutate* category can be linked to patterns such as interactive simulations or side-by-side comparisons, promoting deeper engagement with different scenarios and outcomes. Lastly, *dialogue with* patterns, such as conversational user interfaces, can enhance user-system interaction by creating a narrative of interactions over time.

While these principles provide valuable insights, they remain too abstract to meet the specific needs of DSS designers and lack precise definitions for organizing the flow of information, specifically explanations, between users and systems. To bridge this gap, we organized four workshops, bringing together around 90 experts from various fields, including user experience (UX) design, interaction design, and human-computer interaction (HCI). The workshops had two primary objectives: First, to collaboratively define interactive patterns focused on explanations — what we term explanation stream patterns (ESPs). Second, to identify such patterns that satisfy this definition.

This paper introduces the concept of ESPs and provides illustrative examples of these patterns. By formalizing interaction opportunities through ESPs, we hope to lay the groundwork for future research to systematically quantify their impact on user decision-making. Specifically, we argue that carefully designed ESPs, aligned with users' cognitive processes, could help mitigate the influence of cognitive biases and enhance decision accuracy in DSSs that incorporate XAI.

2. Preliminaries

Research on interactivity in DSSs is inherently interdisciplinary. Information science provides theories for organizing and structuring data [11], while interaction design emphasizes intuitive interfaces that enhance user engagement and usability. HCI examines user-system interactions [12], and XAI focuses on making AI outcomes transparent and interpretable [13,14]. Cognitive psychology links these areas by describing how humans perceive, interpret and act on information, and acknowledging that cognitive biases influence decision-making. Despite this interdisciplinary potential, the integration of these insights remains fragmented, particularly in designing and evaluating interactive systems [15,16,17]. In the remainder of this section, we expand on these aspects.

2.1. Explanations and User Needs

According to the DIKW hierarchy [18,19], which introduces the concepts of 'data', 'information', 'knowledge', and 'wisdom', information organizes data into a meaningful context to support decision-making. In AI-based DSSs, explanations play a critical role in helping users understand how these systems function and arrive at specific conclusions. Explanations provide a specific form of information that clarifies the reasoning behind an AI system's outputs, making its decision-making process transparent and comprehensible to human users.

By bridging the gap between information and actionable knowledge, explanations in XAI act as narratives that enhance user comprehension and trust in AI systems. However, [20] identified a gap between existing XAI methods and user needs. User-centered approaches, such as the development of question banks to guide the design of effective explanations can help fill this gap.

2.2. Patterns

Patterns are an example of such a user-centered approach, being structured reusable solutions to recurring design challenges that enhance usability and aesthetics across interfaces, websites, and programs [7]. Patterns capture best practices within a design domain, offering adaptable solutions that designers can modify based on context. Patterns thus provide flexible solutions that bridge high-level design principles (e.g., "prevent errors") with low-level elements like buttons and text.

Interaction patterns focus on improving how users engage with interfaces, addressing common issues in user flow, navigation, feedback, and control. They help make complex tasks intuitive, reducing cognitive load [7]. We build on interaction patterns and propose ESPs, which are specifically tailored to guide the stream of explanations in DSSs. ESPs focus on how users perceive, interpret, and act on explanations, offering structured

yet adaptable solutions for explanation design. In this study, we focus specifically on patterns related to individual system outcomes — that is, the content and reasoning behind specific decisions or outputs — rather than explaining the system's overall internal logic or general decision-making processes.

2.3. Decision Support Systems

DSSs are developed in response to the understanding that many decision-making scenarios are too complex to be entirely predefined [21,22,23]. Instead of providing fixed solutions, these systems function as collaborative tools, aiding users in analyzing complex information. Numerous works have shown the importance of aligning the advice of a DSS with the cognitive process of users to mitigate biases [24,25,26,27]. By supporting this collaborative process, DSSs enable users to make well-informed decisions.

To align a DSS with the cognitive process of a user, one can, for instance, choose to design an **active** or **passive** system [21,28], where active DSSs proactively offer advice to the user, while passive systems respond only to explicit user requests for analysis or recommendations. When using a **critiquing** mode, feedback is provided only when a user's decision deviates from the advice or recommendations generated by the system. This approach ensures that users receive alerts about potential discrepancies, facilitating better decision making and learning. **Consulting** modes of advice present advice or recommendations even before the decision is made by the user [23] [28].

2.4. Explanation Types

Explanations are a human-centered construct, essential for enhancing transparency and understanding of model predictions, regardless of the model's complexity. The need for explanations remains constant for both simple white-box models like linear regression or complex black-box models like neural networks. However, the methods used to provide explanations vary depending on the model's complexity and the type of information required. Below are some commonly used explanation types as found in [17]:

Feature importance techniques help identify how the input features influence the model's predictions. This is particularly important for black-box models, as understanding the contribution of each feature can be difficult without additional methods. Techniques like SHAP (SHapley Additive exPlanations) and LIME (Local Interpretable Model-agnostic Explanations) are widely used to measure the contribution of each feature to a specific prediction [29]. For white-box models, feature importance is typically straightforward. In linear regression, for example, the coefficients (or weights) assigned to each feature indicate the strength and direction of its influence on the model's output.

Counterfactual explanations highlight how changes to an input could lead to a different outcome [30]. A counterfactual explanation for a loan approval model could indicate that if an applicant's income were $10,000 higher, the loan would be approved, even though all other factors (e.g. credit score, debt level) remain unchanged. This type of explanation is particularly useful for exploring decision boundaries and providing insights that can help users understand what actions they can take to alter a decision.

Similar examples provide insights into how the model behaves for instances that are similar, though not identical, to the one in question. For example, if a model predicts that a loan application will be denied, a similar example might highlight other applicants with

comparable financial profiles who were also denied loans. These explanations focus on patterns in the model's behavior for similar inputs, helping users understand the decision-making process without relying on exact feature matches [31].

Contrastive examples, on the other hand, relate to instances in which the outcome is different compared to the case at hand, highlighting key differences. For example, if an applicant's loan is denied, a contrastive example could show that the loan would be approved if the applicant's credit score were 20 points higher, highlighting how a specific change in the input leads to a different outcome. Note that similar and contrastive examples are related to historical cases, whereas counterfactuals focus on what would happen if specific inputs of the current example were changed.

Rule-based explanations translate a model's decision-making process into human-readable rules that describe how the model arrives at a particular prediction. For example, a rule-based explanation might state, "Approve the loan if the applicant's income is greater than $50,000" or "Deny the loan if the applicant's debt-to-income ratio exceeds 0.4." These rules simplify complex model behavior and make it more accessible to users who may not have a technical background.

The goal of explanation techniques is to provide users with insights into the model's decision-making process. We argue that for each of these explanation types, it is important to model how users perceive, interpret, and act upon the explanations. We provide examples of interactive interventions for the different explanation types in Section 5.

3. Methods

Given our aim to define and identify ESPs, we adopted a qualitative research approach, which is suited for exploring context-dependent phenomena in detail, allowing for the capture of rich insights and nuanced perspectives [32]. We organized four workshops that employed a moderated focus group-like format, facilitating discussions among diverse experts, enabling the collective generation and refinement of ideas.

The first workshop was exploratory in nature and served to establish a consensus-driven definition of ESPs tailored for DSSs. It was organized into three phases: an introduction to relevant concepts, collaborative activities to define and discuss ESPs, and two iterative feedback rounds to refine the definition. After these feedback rounds, consensus was reached among the participants. In three follow-up workshops, equally designed, we identified patterns that align with the final ESP definition.

3.1. Workshop 1: Explanation Stream Patterns Definition

18 researchers (aged 24 to 53, average 36) from a research group on human experience and media design thereby professionally engaged in the study and design of user interfaces attended the workshop. Most had expertise in design disciplines, cognitive science, or human-computer interaction. Their combined experience enabled a multidisciplinary approach to defining ESPs. The session lasted two hours, and participation was voluntary and uncompensated. Participants joined out of personal interest in the topic. (X)AI researchers were not included, as the focus was on interaction and user engagement with explanations, rather than the technical development of explainability methods.

The workshop was aimed at moving from a wide array of aspects related to interaction towards a specific ESP definition. Initially, participants attended a presentation that

introduced DSSs with a human-in-the-loop. Furthermore, several examples of interaction patterns related to progressive disclosure and cognitive forcing were shown to build a shared understanding of what patterns are.

3.1.1. Aspects Related to Controlling Explanation Streams

Following the introduction, participants were asked to indicate important aspects that influence how they engage with and interpret presented information. This started an open discussion on the temporal, spatial and granularity of information presentation, resulting in the keywords depicted in Table 1. The goal of this step was to allow participants to gain a better understanding of aspects influencing decision-making of users in DSSs.

Table 1. Overview of important aspects related to engaging with and interpreting presented information.

Accordion	Chunking	Comprehensibility	Comprehensible	Conversation
Filter	Framing	Hierarchy	Information Tree	Language
Order	Presentation Styles	Progressive Disclosure Disclosure	Scaffolding	Steps
Storytelling Structure	Suggestions	Timing	Transparency	Trustworthiness
Tutorials	UI Capabilities	User Control	User Personalization	Zoom

3.1.2. Initial Definitions and Iterative Refinement

In the next phase, a commonly used definition of Interaction Patterns, along with an initial definition of ESPs was given to the participants:

> *Interaction Patterns* are reusable solutions to common design challenges encountered in the interaction between users and systems. These patterns encapsulate best practices for structuring and facilitating user tasks, ensuring a consistent and intuitive user experience.
>
> *Explanation Stream Patterns* are reusable solutions that govern the stream of explanations between users and a system, particularly in the context of decision support tools. These patterns directly influence how users perceive, interpret, and act on explanations.

To refine the ESP definition, we facilitated several rounds of open discussion and structured feedback. Each iteration enabled participants to collaboratively reflect on and critique the evolving definition. The initial round emphasized open coding of broad conceptualizations of ESPs, while subsequent rounds employed more selective coding to hone in on issues of clarity and contextual applicability. This recursive process of refinement continued until thematic saturation was reached and participants expressed consensus. The final operationalised definition is presented in Section 4.

3.1.3. Workshop 2,3,4: Brainstorming on Explanation Stream Patterns

Three workshops were conducted to identify patterns aligned with the final ESP definition. Each three-hour session included around 24 of the total of 72 data-driven design master's students with backgrounds in interaction design, UX, and data science. The

group was aged early 20s to late 30s. Participants were not compensated and worked in small groups of four to six to independently identify ESPs. In phase one, groups analyzed ten provided XAI articles, followed by an organizer-led discussion to share and compare findings. In the second phase, groups independently explored additional literature and analyzed other recommender or decision-support systems of their choice to identify more ESPs followed again with a final group discussion. Identified patterns included structured data visualization, adaptive information presentation, and real-time feedback mechanisms — each reflecting different approaches to conveying information to users. The aim was to compile an initial list of ESPs and categorize them by intended goals. These patterns are listed in Table 2.

4. Results

This section provides the results of the workshops. First, the resulting definition is given, after which the identified ESPs are described.

4.1. Definition Explanation Stream Patterns

Through the iterative process described above, consensus was reached on a definition:

> *Explanation Stream Patterns* are reusable solutions that govern the stream of explanations between users and systems, specifically in the context of decision support tools. These patterns aim to influence how users perceive, interpret, and act on explanations.

4.2. Categories of Explanation Stream Patterns

The workshop resulted in twelve ESPs for managing the flow of explanations between users and systems, as presented in Table 2. These patterns address diverse challenges in explanation design and vary in their level of granularity. Some patterns, such as Progressive Disclosure and Cognitive Forcing, provide high-level strategies for structuring explanations and managing complexity, while others, like Tooltips and Mouse-Over or Breadcrumbs, focus on more specific interface components that enhance user interaction. To better understand and systematically organize these emerging patterns, we conducted thematic analysis using axial coding to identify key categories. This process led to the development of several categories, each reflecting distinct interactivity goals.

- Progressive Disclosure patterns focus on controlling how much information is shown and when it is revealed, incrementally providing detail at the right time. These patterns structure complexity by breaking explanations into manageable layers, allowing users to access deeper information on demand. Patterns such as *Accordion*, *Breadcrumbs*, and *Tooltips and Mouse-Over* align with this category. To date, limited research has been conducted on the effectiveness of progressive disclosure techniques, though existing studies suggest that these techniques significantly contribute to enhancing transparency in systems [33,8].

Table 2. Explanation Stream Patterns distilled from the workshops.

Pattern	Description
Progressive Disclosure	Information is presented progressively, either sequentially, hierarchically, or based on user interaction. It is disclosed gradually in response to user actions or decisions.
Cognitive Forcing	Strategies that involve prompting users or the system to engage in specific actions that guide decision-making. This includes user-based strategies such as ranking features, modifying inputs, or exploring counterfactuals, and system-based strategies such as limiting navigation, delaying information, or requiring confirmation before proceeding.
Zooming and Scaling	Allows users to zoom in for more detailed views or zoom out for broader context, facilitating better comprehension of information.
Filtering	Enables users to narrow down displayed information by selecting specific criteria, aiding focus on relevant information.
Tooltips and Mouse-Over	Provides contextual information when the user hovers over UI elements, offering immediate, supplementary information.
Breadcrumbs	Displays the user's navigation path within the system, helping them track their position and navigate backwards.
Accordion	Pattern that expands or collapses sections of content, allowing users to explore or hide information as needed.
User Inputs	Allows users to input predefined criteria or preferences, enabling them to customize both the system's behavior and outputs, as well as modify the inputs provided to the model.
Local Proximity	Group related information or objects closely together.
Context Bundling	Presents related information together in a single context.
Highlighting Important Regions/Features	Uses visual emphasis to direct the user's attention to critical features or information within the interface.
Defaults	Bundle together actions or suggestions for the users.

- Cognitive Forcing patterns allow users to engage cognitively with the system by prompting them to question or reflect on its outputs, promoting critical thinking. Patterns like *Highlighting Important Regions/Features* and *Defaults* fit within this category as they encourage users to critically evaluate specific aspects of the explanation or confirm their decisions. *User Inputs* also align with this category by prompting users to engage actively in customizing inputs or selecting relevant features before proceeding. Numerous research articles have been published on cognitive forcing techniques [34,35,36].
- Behavioral Nudging patterns aim at steering user behavior, including guiding exploration or promoting effective interaction with explanations. Non-coercive nudging guides individuals toward better choices while leaving them with the freedom to choose differently if they wish. Coercive nudging, on the other hand, forces behavior or imposes penalties for certain choices, creating more pressure on the individual to comply with the desired behavior. Patterns such as *Highlighting Important Regions/Features*, *Defaults*, and *Filtering* are effective nudging mechanisms. These patterns guide user attention and behavior by emphasizing critical or important elements, provide options, or narrowing the displayed information to reduce cognitive overload. Limited research has been conducted on nudging in AI-assisted decision-making, with some examples provided in [36,37].

- User-Driven Customization patterns allow users to explicitly configure their preferred explanation formats, types, or levels of detail. View-level customizations enable users to adjust how explanations are presented (e.g., visual vs. textual formats, detail depth). Additionally, model-level customizations allow users to modify the operation of the system or model itself, such as adjusting inputs, altering parameters, or triggering updates to the model's behavior. Patterns like *User Inputs* and *Filtering* provide users tools to explicitly tailor their explanation needs.
- System-Driven Customization patterns infer the user's understanding and dynamically adapt explanations to align with their mental model. Patterns such as *Local Proximity* and *Context Bundling* support this category by dynamically organizing information based on inferred user preferences. These patterns improve system adaptability by presenting related information in cohesive groupings, streamlining user understanding without requiring explicit input. Many research articles exists related to personalized XAI [38,39,40], mostly accounting for different user groups rather than taking into account user characteristics or adaptations of the explanations themselves.

These categories provide a comprehensive overview for understanding how explanation streams can be managed, presented, and interacted with within DSSs.

5. Explanation Types and Explanation Stream Patterns

ESPs can help in structuring and organizing the flow of explanations based on the explanation type. This can be achieved by integrating specific explanation types with suitable interaction interventions. Several examples are listed in this section for each explanation type. It is important to highlight that research on dynamic, interactive explanations remains largely overlooked. The majority of existing studies focus on static explanations, offering limited or no opportunities for user interaction. As such, it is generally not possible to reference examples from existing studies. We propose several interaction strategies for engaging with explanations. Future work should focus on building a comprehensive pattern library related to ways to interact with explanations.

5.1. Feature Importance

Feature importance explanations can be communicated by using **progressive disclosure**. Initially, showing only the most important features (e.g., the top 3 or 5) allows users to focus on key information first [41]. These features can be presented in numerous formats, such as a list that can be extended upon request, commonly referred to as an accordion pattern. This approach helps users process information incrementally, which influences how users perceive and interpret the information.

To further enhance engagement, **cognitive forcing** can be applied by making users expand the list themselves, which encourages them to think critically about the most important features before seeing the entire set. In addition, subtle forms of **behavioral nudging** — such as visually emphasizing the top features or pre-selecting recommended ones — can steer user attention without restricting their autonomy. For personalization, **user-driven customization** can allow users to prioritize or hide specific features, while **system-driven customization** can adjust the list dynamically based on historical user behavior, for example, always presenting the most relevant features first.

5.2. Counterfactuals

Counterfactual scenarios can be communicated by starting with simple explanations illustrating the effect of a single feature change, then expanding to more complex scenarios involving multiple changes. Using the accordion pattern, users can explore a single counterfactual scenario initially and expand it to reveal more counterfactuals as needed.

Progressive disclosure ensures that users are not overloaded with too many possibilities at once, presenting them with simple changes and then revealing more complex alternatives step-by-step. By forcing users to expand the scenarios themselves, **cognitive forcing** ensures that they actively engage with the potential impacts of each change. Additionally, **nudging** can subtly suggest the most impactful counterfactuals, helping users focus on the changes that could lead to better outcomes. To make the explanation more tailored, **user-driven customization** allows users to select which aspects of the counterfactuals they want to explore, while **system-driven customization** dynamically adjusts the scenarios based on the user's past behavior or current context.

5.3. Similar/Contrastive Examples

To make comparisons between similar and contrasting examples, **progressive disclosure** can be effective. Start by presenting a basic, similar example, and then allow users to expand the explanation to view more contrasting examples. This helps prevent information overload while still offering the opportunity to explore different outcomes in depth.

As with other explanation types, **cognitive forcing** can be employed by gradually increasing the contrast between examples, prompting users to reflect on the differences and understand the impact of each decision. **Nudging** can be applied by highlighting examples that lead to the desired outcome, guiding the user towards the most relevant or beneficial scenarios. **User-driven customization** also plays a key role here, allowing users to search for or select the types of examples they wish to see, ensuring that the examples align with their needs. Similarly, **system-driven customization** can recommend the most pertinent examples based on the user's interactions and preferences.

5.4. Rule-Based Explanations

For rule-based explanations, such as decision trees or decision rules, **progressive disclosure** can be used. An example could be to first present high-level rules after which users can then expand to reveal more detailed information about the decision tree or specific decision rules. This layered approach reduces cognitive load while still providing the depth required for full understanding.

Cognitive forcing can be used here as well, ensuring that users engage with the logic of each rule step-by-step, rather than overwhelming them with the entire tree or set of rules at once. **Nudging** can help guide users by highlighting key decision points or rules that significantly impact the outcome, subtly steering them towards the most relevant parts of the logic. Additionally, **user-driven customization** allows users to select which parts of the decision tree or which rules they want to explore, making the explanation more relevant to their interests. **System-driven customization** might dynamically adapt the explanation based on the user's past interactions, showing the most relevant rules based on their preferences or decision history.

6. Discussion

ESPs provide practical tools for structuring explanation streams in DSSs. By formalizing ESPs as reusable solutions that influence user interaction with explanations, a shared design space emerges that bridges the gap between interaction designers and developers.

6.1. Foundation for Designing Explanation Streams

The ESP framework provides a theoretical and practical foundation for analyzing and designing explanation flows within DSSs. By dissecting explanation streams into specific patterns, ESPs formalize aspects of information exchange that were previously unstructured or implicitly assumed. This approach may inspire new research into the impacts of different ESPs, particularly regarding how users perceive, interpret, and act on explanations. Insights gained could lead to the refinement of ESP categories or even new patterns, further contributing to the discipline of decision support. By providing these theoretical building blocks, ESPs create a pathway for more structured research into how to effectively design and facilitate interactions between users and AI systems through explanations. We envision a comprehensive pattern library that could facilitate the above, potentially steered by the explanation needs as identified in [20].

6.2. Common Design Space for Decision Support System Development

The definition and categorization of ESPs create a standardized design space where designers and developers can align on explanation flows and interaction within DSSs. This shared framework allows for clear communication of design intents, such as when to deploy static or dynamic flows and when to guide or allow for exploratory engagement. ESPs could help prevent misunderstandings that might otherwise arise from abstract goals, such as ”improving decision quality” or ”reducing cognitive bias.” Instead, these goals could be tied to specific patterns, facilitating a concrete approach for both design and implementation.

6.3. Bias Mitigation and Alignment with Cognitive Needs

ESPs offer opportunities to model interactivity in DSS designs, aligning them with human cognitive needs and potentially improving decision-making effectiveness. A wide range of ESPs can be developed and evaluated in user-specific contexts to assess their impact on human-AI decision-making. Future research should explore to what extent ESPs mitigate cognitive biases and further quantify their influence on the quality of user decision-making in various human-AI decision support contexts.

6.4. Decision-Context and User Needs

In this paper, we formalized interactivity through ESPs. While not covered in this paper, it is important to note that explanations are dependent on the decision context, which is shaped by factors such as high stakes, irreversibility, and time pressure [42,43,44]. Such factors significantly influence explanation design by tailoring patterns such as granularity and depth to align with the urgency, importance, and consequences of the decision. For example, in time-critical decisions, the system may prioritize concise and actionable

explanations, while in high-stakes or irreversible decisions, it may provide detailed explanations that include ethical and legal justifications to support accountability and trust.

Similarly, user needs play a crucial role in explanation design by ensuring that the information presented aligns with the user's role, goals, and level of expertise [45,46]. Explanations must be tailored to address specific needs, whether the user is a domain expert or a layperson, and whether their focus is task-oriented or exploratory. For instance, a clinician may require explanations to design personalized speech treatment plans that meet the specific needs of each patient, while patients may use explanations to stay motivated throughout treatment [47,48].

Together, these contexts guide the customization of ESPs, ensuring they are both situationally appropriate and user-centered. Future research should build on these foundations by considering these contexts in the design of ESPs to ensure their relevance and effectiveness.

7. Conclusion

In this research, we established a definition for ESPs and provided several examples of ways to interact with explanations. The goal of these ESPs is to provide a more formalized approach to designing interactive explanations. ESPs address the current lack of standardization in evaluating and designing explanations, which hinders efforts to communicate explanations effectively.

In the workshops, we identified several ESPs and distilled them into five categories that shape the intent by which explanations are communicated. Future work should involve a comprehensive analysis of the literature surrounding ESPs to uncover existing frameworks, methodologies, and findings that can enhance our understanding of how to communicate explanations in a structured way for DSSs. A mapping study could be an interesting perspective to see how ESPs connect to specific explanation needs as found in [20]. Furthermore, developing a pattern library could serve as a practical resource for standardizing and applying ESPs across various contexts.

Future research is essential to investigate the effectiveness of ESPs in mitigating cognitive biases. While the proposed patterns offer a promising research perspective, empirical studies are needed to quantify their impact on decision-making outcomes. Understanding how ESPs influence human-AI decision-making accuracy will be crucial in validating their practical use. Specifically, research should focus on assessing changes in decision quality, user behavior, and how interactivity with explanations in DSSs can help mitigate cognitive biases.

Moreover, we suggest exploring the effectiveness of ESPs in different decision-making contexts/domains. Different domains present unique challenges and user needs [6]. Tailoring ESPs to address the specific cognitive requirements and contextual factors within these sectors can enhance their effectiveness. For instance, the decision-making processes in healthcare may demand different ESPs compared to those used in telecommunications due to considerations such as differing stakes, time constraints for decisions, stressful conditions, and the complexity of the information involved [49,50]. Context-dependent ESPs could be employed to adjust the explanation stream based on situational factors, such as presenting explanations differently when the model's confidence is low, in time-sensitive scenarios, or during irreversible decision-making processes.

References

[1] Veale M, Van Kleek M, Binns R. Fairness and accountability design needs for algorithmic support in high-stakes public sector decision-making. In: Proceedings of the 2018 chi conference on human factors in computing systems; 2018. p. 1-14.

[2] Musen MA, Middleton B, Greenes RA. Clinical decision-support systems. In: Biomedical informatics: computer applications in health care and biomedicine. Springer; 2021. p. 795-840.

[3] Angehrn AA, Lüthi HJ. Intelligent decision support systems: a visual interactive approach. Interfaces. 1990;20(6):17-28.

[4] Antoniadi AM, Du Y, Guendouz Y, Wei L, Mazo C, Becker BA, et al. Current challenges and future opportunities for XAI in machine learning-based clinical decision support systems: a systematic review. Applied Sciences. 2021;11(11):5088.

[5] Ali S, Abuhmed T, El-Sappagh S, Muhammad K, Alonso-Moral JM, Confalonieri R, et al. Explainable Artificial Intelligence (XAI): What we know and what is left to attain Trustworthy Artificial Intelligence. Information fusion. 2023;99:101805.

[6] Chromik M, Butz A. Human-XAI interaction: a review and design principles for explanation user interfaces. In: Human-Computer Interaction–INTERACT 2021: 18th IFIP TC 13 International Conference, Bari, Italy, August 30–September 3, 2021, Proceedings, Part II 18. Springer; 2021. p. 619-40.

[7] Tidwell J. Designing interfaces: Patterns for effective interaction design. " O'Reilly Media, Inc."; 2010.

[8] Springer A, Whittaker S. Progressive disclosure: empirically motivated approaches to designing effective transparency. In: Proceedings of the 24th international conference on intelligent user interfaces; 2019. p. 107-20.

[9] Van Brussel S, Timmermans M, Verkoeijen P, Paas F. 'Consider the Opposite'–Effects of elaborative feedback and correct answer feedback on reducing confirmation bias–A pre-registered study. Contemporary Educational Psychology. 2020;60:101844.

[10] Bertrand A, Viard T, Belloum R, Eagan JR, Maxwell W. On selective, mutable and dialogic XAI: A review of what users say about different types of interactive explanations. In: Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems; 2023. p. 1-21.

[11] Borko H. Information science: what is it? American documentation. 1968;19(1):3-5.

[12] Preece J, Rogers Y, Sharp H, Benyon D, Holland S, Carey T. Human-computer interaction. Addison-Wesley Longman Ltd.; 1994.

[13] Xu F, Uszkoreit H, Du Y, Fan W, Zhao D, Zhu J. Explainable AI: A brief survey on history, research areas, approaches and challenges. In: Natural language processing and Chinese computing: 8th cCF international conference, NLPCC 2019, dunhuang, China, October 9–14, 2019, proceedings, part II 8. Springer; 2019. p. 563-74.

[14] Gunning D, Stefik M, Choi J, Miller T, Stumpf S, Yang GZ. XAI—Explainable artificial intelligence. Science robotics. 2019;4(37):eaay7120.

[15] Lopes P, Silva E, Braga C, Oliveira T, Rosado L. XAI systems evaluation: A review of human and computer-centred methods. Applied Sciences. 2022;12(19):9423.

[16] Benchekroun O, Rahimi A, Zhang Q, Kodliuk T. The need for standardized explainability. arXiv preprint arXiv:201011273. 2020.

[17] Kim J, Maathuis H, Sent D. Human-centered evaluation of explainable AI applications: a systematic review. Frontiers in Artificial Intelligence. 2024;7. Available from: https://www.frontiersin.org/journals/artificial-intelligence/articles/10.3389/frai.2024.1456486.

[18] Ackoff RL. From data to wisdom. Journal of applied systems analysis. 1989;16(1):3-9.

[19] Rowley J. The wisdom hierarchy: representations of the DIKW hierarchy. Journal of information science. 2007;33(2):163-80.

[20] Liao QV, Gruen D, Miller S. Questioning the AI: informing design practices for explainable AI user experiences. In: Proceedings of the 2020 CHI conference on human factors in computing systems; 2020. p. 1-15.

[21] Medlock S, Wyatt JC, Patel VL, Shortliffe EH, Abu-Hanna A. Modeling information flows in clinical decision support: key insights for enhancing system effectiveness. Journal of the American Medical Informatics Association. 2016;23(5):1001-6.

[22] Guerlain SA, Smith PJ, Obradovich JH, Rudmann S, Strohm P, Smith JW, et al. Interactive critiquing as a form of decision support: An empirical evaluation. Human factors. 1999;41(1):72-89.

[23] Irandoust H. Critiquing systems for decision support. DRDC Valcartier (TR 2003-321) Valcartier, Quebec: Defence R&D Canada–Valcartier, Department of National Defence DRDC Toronto TR.

2006;259:35.
[24] Fürnkranz J, Kliegr T, Paulheim H. On cognitive preferences and the plausibility of rule-based models. Machine Learning. 2020;109(4):853-98.
[25] Chromik M, Eiband M, Buchner F, Krüger A, Butz A. I think i get your point, AI! the illusion of explanatory depth in explainable AI. In: Proceedings of the 26th International Conference on Intelligent User Interfaces; 2021. p. 307-17.
[26] Nourani M, Roy C, Block JE, Honeycutt DR, Rahman T, Ragan E, et al. Anchoring Bias Affects Mental Model Formation and User Reliance in Explainable AI Systems. In: Proceedings of the 26th International Conference on Intelligent User Interfaces. IUI '21. New York, NY, USA: Association for Computing Machinery; 2021. p. 340–350.
[27] Wang D, Yang Q, Abdul A, Lim BY. Designing Theory-Driven User-Centric Explainable AI. In: Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems. CHI '19. New York, NY, USA: Association for Computing Machinery; 2019. p. 1–15. Available from: https://doi.org/10.1145/3290605.3300831.
[28] Kubben P, Dumontier M, Dekker A. Fundamentals of Clinical Data Science. Springer Nature; 2019.
[29] Lundberg S. A unified approach to interpreting model predictions. arXiv preprint arXiv:170507874. 2017.
[30] Wachter S, Mittelstadt B, Russell C. Counterfactual explanations without opening the black box: Automated decisions and the GDPR. Harv JL & Tech. 2017;31:841.
[31] Miller T. Explanation in artificial intelligence: Insights from the social sciences. Artificial Intelligence. 2019;267:1-38. Available from: https://www.sciencedirect.com/science/article/pii/S0004370218305988.
[32] Pedersen MA. Editorial introduction: Towards a machinic anthropology. Big Data & Society. 2023;10(1):20539517231153803. Available from: https://doi.org/10.1177/20539517231153803.
[33] Muralidhar D. Operationalizing Selective Transparency Using Progressive Disclosure in Artificial Intelligence Clinical Diagnosis Systems. Available at SSRN 5062641. 2024.
[34] Buçinca Z, Malaya MB, Gajos KZ. To trust or to think: cognitive forcing functions can reduce overreliance on AI in AI-assisted decision-making. Proceedings of the ACM on Human-computer Interaction. 2021;5(CSCW1):1-21.
[35] Simkute A, Luger E, Evans M, Jones R. Experts in the shadow of algorithmic systems: Exploring intelligibility in a decision-making context. In: Companion Publication of the 2020 ACM Designing Interactive Systems Conference; 2020. p. 263-8.
[36] Naiseh M, Al-Mansoori RS, Al-Thani D, Jiang N, Ali R. Nudging through friction: an approach for calibrating trust in explainable AI. In: 2021 8th International Conference on Behavioral and Social Computing (BESC). IEEE; 2021. p. 1-5.
[37] Erdeniz SP, Tran TNT, Felfernig A, Lubos S, Schrempf M, Kramer D, et al. Employing nudge theory and persuasive principles with explainable ai in clinical decision support. In: 2023 IEEE International Conference on Bioinformatics and Biomedicine (BIBM). IEEE; 2023. p. 2983-9.
[38] Conati C, Barral O, Putnam V, Rieger L. Toward personalized XAI: A case study in intelligent tutoring systems. Artificial Intelligence. 2021;298:103503. Available from: https://www.sciencedirect.com/science/article/pii/S0004370221000540.
[39] Subramanian HV, Canfield C, Shank DB. Designing explainable AI to improve human-AI team performance: A medical stakeholder-driven scoping review. Artificial Intelligence in Medicine. 2024;149:102780. Available from: https://www.sciencedirect.com/science/article/pii/S0933365724000228.
[40] Saeed W, Omlin C. Explainable AI (XAI): A systematic meta-survey of current challenges and future opportunities. Knowledge-Based Systems. 2023;263:110273. Available from: https://www.sciencedirect.com/science/article/pii/S0950705123000230.
[41] Ghai B, Liao QV, Zhang Y, Bellamy R, Mueller K. Explainable active learning (xal) toward ai explanations as interfaces for machine teachers. Proceedings of the ACM on Human-Computer Interaction. 2021;4(CSCW3):1-28.
[42] Kunreuther H, Meyer R, Zeckhauser R, Slovic P, Schwartz B, Schade C, et al. High stakes decision making: Normative, descriptive and prescriptive considerations. Marketing Letters. 2002;13:259-68.
[43] Pauwels N, Van De Walle B, Hardeman F, Soudan K. The implications of irreversibility in emergency response decisions. Theory and Decision. 2000;49(1):25-51.

[44] Wu CM, Schulz E, Pleskac TJ, Speekenbrink M. Time pressure changes how people explore and respond to uncertainty. Scientific reports. 2022;12(1):4122.

[45] Langer M, Oster D, Speith T, Hermanns H, Kästner L, Schmidt E, et al. What do we want from Explainable Artificial Intelligence (XAI)?–A stakeholder perspective on XAI and a conceptual model guiding interdisciplinary XAI research. Artificial Intelligence. 2021;296:103473.

[46] Van Den Berg M, Kuiper O, Van Der Haas Y, Gerlings J, Sent D, Leijnen S. A conceptual model for implementing explainable ai by design: Results of an empirical study. In: HHAI 2023: Augmenting Human Intellect. IOS Press; 2023. p. 60-73.

[47] Kim M, Kim S, Kim J, Song TJ, Kim Y. Do stakeholder needs differ? - Designing stakeholder-tailored Explainable Artificial Intelligence (XAI) interfaces. International Journal of Human-Computer Studies. 2024;181:103160. Available from: https://www.sciencedirect.com/science/article/pii/S1071581923001696.

[48] Langer M, Oster D, Speith T, Hermanns H, Kästner L, Schmidt E, et al. What do we want from Explainable Artificial Intelligence (XAI)? – A stakeholder perspective on XAI and a conceptual model guiding interdisciplinary XAI research. Artificial Intelligence. 2021;296:103473. Available from: https://www.sciencedirect.com/science/article/pii/S0004370221000242.

[49] He X, Hong Y, Zheng X, Zhang Y. What are the users' needs? Design of a user-centered explainable artificial intelligence diagnostic system. International Journal of Human–Computer Interaction. 2023;39(7):1519-42.

[50] Phillips-Wren G, Power DJ, Mora M. Cognitive bias, decision styles, and risk attitudes in decision making and DSS. Journal of Decision Systems. 2019;28(2):63-6.